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Aerodynamic Concept Evaluation of Formula Student Side Structure

GABRIEL ÅGREN



Aerodynamic Concept Evaluation of Formula Student Side Structure
Gabriel Ågren <gaagren@kth.se>

Supervisor: Mikael Nybacka
Examiner: Annika Stensson Trigell

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School of Engineering Science
KTH Royal Institute of Technology

Abstract

Formula Student is a global engineering competition where university students collaborate to design, construct, and race formula-style cars. Aerodynamics is one aspect in the vehicle design that can improve on-track performance by increasing cornering and straight-line speed.

To improve the aerodynamics of KTH Formula Student's DeV18 vehicle, the side structure is being redesigned. The current model, DeV17, features an underperforming tunnel-based side structure. To address this issue, this had the goal to investigate a new multi-element wing design that utilizes ground effect.

The design study of the DeV18 vehicle is conducted using Siemens NX 2212 for 3D modelling and Simcenter Star-CCM+ 17.06.008-R8 for airflow simulations. To quickly investigate certain design parameters effect on the results, Design Manager Project inside Simcenter Star-CCM+ is used.

The resulting side structure produces a total of 26 N of downforce and 6 N of drag at 40 kph, more than twice that of DeV17's side structure while also producing less drag. Although this significant improvement compared to DeV17, it is believed that further increases in performance are necessary to compete with top teams. By using a more sophisticated method to optimize the multi-element wing, such as adjoint optimization, the concept could be improved. However, the overall potential of the concept is still considered too limited to achieve the desired performance goals, which is why it will no longer be investigated further.

Sammanfattning

Formula Student är en global ingenjörstävling där universitetsstudenter samarbetar för att designa, konstruera och tävla med formelbilar. Aerodynamik är en aspekt i fordonsdesignen som kan förbättra prestandan på banan genom att öka kurvastigheter och hastigheter på raksträckor.

För att förbättra aerodynamiken på KTH Formula Students DeV18-fordon genomgår sidostrukturen en omkonstruktion. Den nuvarande bilen, DeV17, har idag en underpresterande tunnelbaserad sidostruktur och för att förbättra detta till nästa bil undersöks, i den här rapporten, en flerdelad vingdesign som utnyttjar markeffekt.

Designstudien av DeV18-fordonet genomförs med hjälp av Siemens NX 2212 för 3D-modellering och Simcenter Star-CCM+ 17.06.008-R8 för luftflödessimuleringar. För att snabbt kunna undersöka vissa designparametrars effekt på resultatet används Design Manager Project inuti Simcenter Star-CCM+.

Den resulterande sidostrukturen producerar en total negativ lyftkraft på 26 N och 6 N luftmotstånd vid 40 km/h, vilket är mer än dubbelt så mycket som DeV17:s sidostruktur samtidigt som den producerar mindre luftmotstånd. Även om denna förbättring, jämfört med DeV17, är av betydande storlek, tros det att ytterligare ökningar i prestanda är nödvändigt för att kunna konkurrera med topplagen. Genom att använda en mer sofistikerad metod för att optimera den flerdelade vingen, såsom adjointoptimering, kan konceptet förbättras. Men den övergripande potentialen för konceptet anses fortfarande vara för liten för att uppnå de önskade prestandamålen, vilket är varför det här konceptet inte längre kommer att undersökas vidare.

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1 Introduction

1.1 Background

1.1.1 Formula Student

Formula Student is widely recognized as the largest educational engineering competition globally. It receives substantial support from various industries and has a primary objective of fostering the growth and development of enterprising and innovative engineers, as well as promoting careers in engineering among young people.

The competition poses a challenge to its participating teams, requiring them to integrate intensive experiences in design and manufacturing into their education. Additionally, teams are expected to consider the economic and marketing aspects of the automotive industry and to assume the role of a manufacturer, tasked with developing a prototype for production evaluation - a common assignment for engineers.

The target audience for Formula Student is non-professional racers for whom the race car must exhibit exceptional driving dynamics and be reliable, dependable, and economically feasible. The market value of the car is also influenced by factors such as aesthetics, comfort, and the utilization of readily available, standard components.

Formula Student primarily focuses on the innovation and problem-solving skills required of engineers rather than the motorsport aspect of the competition. The organization is heavily influenced by the supporting industries, reflecting the need for skilled engineers capable of addressing modern challenges such as electrification and autonomous transportation of vehicles. The development of electric vehicles with driverless capabilities is, therefore, a highly relevant and pertinent task. [1]

1.1.2 Race car Aerodynamics

Race car aerodynamics is the study of how the shape and design of a race car affects its movement through the air. The goal of race car aerodynamics is to minimize air resistance and increase downforce and stability, all of which can improve the car's performance on the track.

By changing the shape of the car and implementing aerodynamic devices, such as a rear wing or flat floor, the airflow around the vehicle can be controlled to either create or decrease pressure differences in desired places, reducing the air resistance of the vehicle and/or forcing the car into the ground increasing grip levels.

Ever since understanding how crucial the shape of the car and its aerodynamics devices is for top speeds and the vehicles' ability to carry speed through corners it has rose to become a vocal point of development for racing teams and automotive manufacturers. For instance, F1 teams invest as much as 20 % of their total budget in aerodynamics. [2]

1.1.3 Formula Student and Aerodynamics

In recent years, aerodynamics in formula student has become a big part of the sport and is almost becoming a defining factor when describing a FS-car, see Figure 1.



Figure 1. TUFast Racing Teams car from the 2022 season. [3]

Aerodynamics on Formula Student vehicles has historically been quite a heated debate. It is not unusual for people inside the Formula Student community to argue that aerodynamic elements on an FS car will not increase overall performance. This is often based on the low velocities the track allows for, resulting in low downforce levels, which are not enough to counteract the negatives of implementing an aerodynamics package, such as weight, a higher centre of gravity, and the time spent developing and manufacturing the package that could have been spent elsewhere. However, it is widely agreed that if an aerodynamic package has been well-developed from a vehicle dynamics perspective and has been produced with high precision, aerodynamics can give a net increase in performance. In the autocross event at FSG 2022 in the electric category, the top 15 teams had an aerodynamics package. [4]

1.2 Purpose

By aerodynamically optimizing the side structure of KTH Formula Student's vehicle, DeV18, the team improves its ability to perform well at competitions.

1.3 Limitations

The study will not include a full evaluation of the design concept. Instead, it will aim to gain an understanding of the design concept in question and assess its potential. To conserve computational power, the study will exclude other aerodynamic elements of the car that doesn't directly affect the performance of the side structure. Sensitivity studies such as pitch, roll, and yaw studies, which are necessary for a thorough understanding of the design, will not be included since those studies aren't considered necessary for understanding the potential of the concept. Lastly, the study will not verify the correlation between reality and

simulation. The simulations performed will be purely based on previous correlation studies conducted by the team and literature.

1.4 Sustainability

Understanding the flow field around a moving vehicle and optimizing its shape can have a significant impact on sustainability. Aerodynamic drag on a vehicle can account for a large portion of its total energy consumption. Therefore, understanding the flow field around a vehicle and knowing how to control and change it can lead to a reduction in energy consumption. By reducing the turbulent wake behind the vehicle, noise pollution can also be reduced.

2 Theory

2.1 Race car Aerodynamics

Race car aerodynamics is the study of how airflow interacts with a race car to improve its performance. By using aerodynamic elements to manipulate the airflow around the car, downforce can be increased and drag decreased which allows for higher cornering speeds and top speeds.

2.1.1 Ground Effect

Ground effect in terms of race cars is a phenomenon that occurs when the car moving. The air traveling underneath the car accelerates resulting in a low-pressure zone under neath the car, pushing the car into the ground. Through different designs the ground effect can be increased or decreased, see Figure 2.

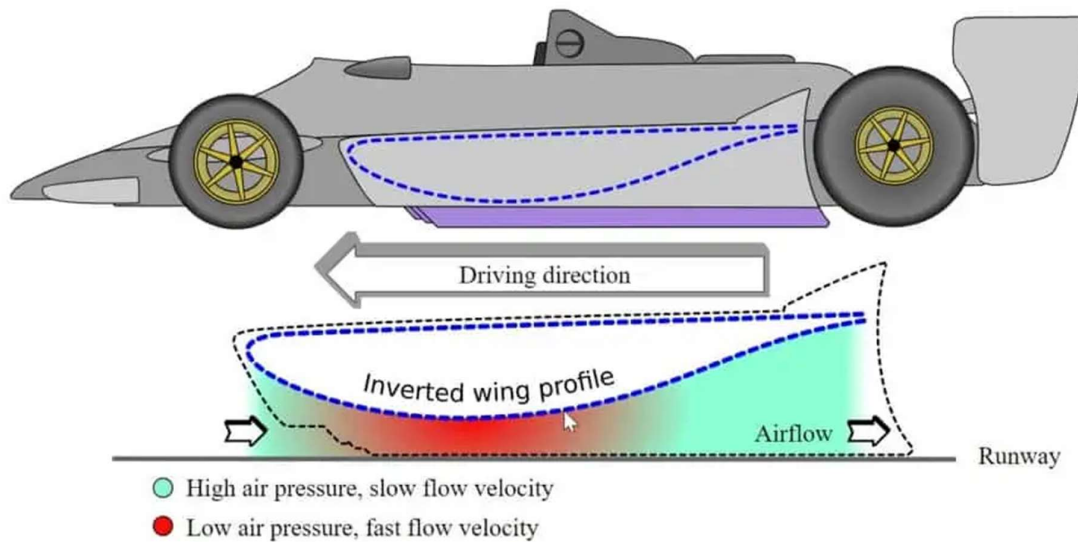


Figure 2. Picture of how ground effect was used in the 1980's. [5]

2.1.2 Multi-element Wings

Multi-element wings are commonly used on race cars to improve their aerodynamic performance. These wings consist of multiple wing elements stacked on top of each other, with each element generating downforce to increase traction and stability. The main element

creates the majority of the downforce, while the smaller elements or flaps work in conjunction with the main element to fine-tune the airflow and provide additional downforce at specific speeds or angles of attack. The design and positioning of these elements are critical to get the them to work well together, improving the performance of each other.

Recommended gap values are between one and four percent of the total chord length and one and six percent for overlap values, with an angle of attack for Flap 1 between 25 and 30 degrees and 30 and 70 for additional flaps. [6]

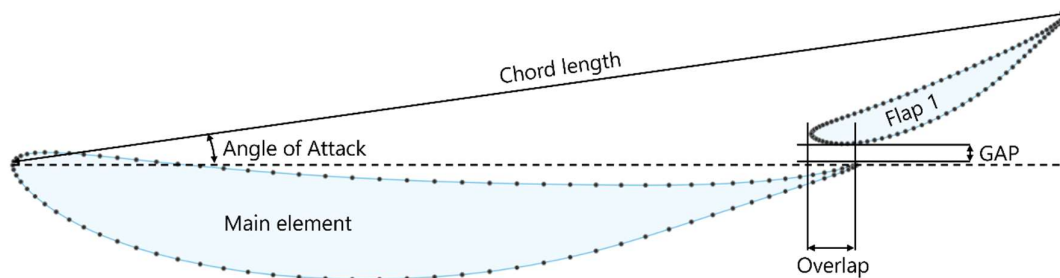


Figure 3. A multi-element wing with the definition of multi-element wing chord length, angle of attack, gap, and overlap.

2.1.3 Tyre Wake and Squash

In racing, the term "tire wake" refers to the turbulent air flow that is generated behind the tires of a race car as it moves at high speeds. This wake can have a negative impact on the car's aerodynamics, as it creates low energy airflow that can reduce performance of aerodynamic elements.

The tire wake is created as air flows over the surface of the tire and separates from it. This separation causes turbulence in the air flow, creating a wake of swirling, chaotic air behind the tire. Tire squash on the other hand is created as airflow gets pulled down with the frontward facing tire face and gets squashed between the tire and ground to eventually spill on either side of the tire. This airflow also has low energy and can reduce the performance of aerodynamic elements.

2.2 Fluid Mechanics

2.2.1 Boundary Layer

When there is relative motion between a fluid and object, a small region on the object's surface is created where velocity, pressure, and temperature changes due to the interaction between the two. This region is known as the boundary layer and is characterized by a gradual decrease in velocity as the distance to the object surface decreases.

2.2.2 Streamlines

Streamlines are lines that shows the direction of fluid flow at any given location within a flow field. At each point in the flow field, streamlines are always tangent to the velocity vector,

and they never cross each other. Studying streamlines can give an understanding of how fluid flows around a body.

2.2.3 Wall Shear Stress

Wall shear stress is the force per unit area that is put, by a fluid, on an object surface. This stress is created by the interaction between the fluid and the surface and studying wall shear stress on an object surface can be used as an indicator of when airflow separates from the surface.

2.2.4 Q-criterion

Q-criterion is a mathematical tool used to visualize behaviour of the flow, more specifically, vortices but also overall structure of the flow field. This works by analysing when fluid particles are stretched out into a vortex.

2.3 Computational Fluid Dynamics (CFD)

2.3.1 Navier-Stokes Equations

The Navier-Stokes equations are a set of partial differential equations that describe the motion of fluids. The equations describe the fundamental relationship between the velocity, pressure, density, and temperature of a fluid. They are used to calculate how a fluid will move and behave under different conditions, such as changes in temperature, pressure, or the presence of external forces like gravity.

The differential form of the continuity equation is expressed in Equation 1, while the energy equation deals with the exchange between different forms of energy, such as kinetic and potential energy. In the case of Formula Student vehicles, the Navier-Stokes equations can be simplified by assuming incompressible flow and neglecting the energy equation due to the negligible temperature gradient. As a result, the continuity and momentum equations can be simplified to the incompressible version of the momentum equation and we can apprehend the set of partial differential equations known as the Navier-Stokes equations for incompressible flow, Equation 2.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

Equation 1

$$\begin{cases} \nabla \cdot \mathbf{u} = 0 \\ \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f} \end{cases}$$

Equation 2

Where $\mathbf{u} = u_x, u_y, u_z$ is the fluids flow velocity,

ρ is the fluid density,

$\mathbf{f} = f_x, f_y, f_z$ are body forces acting on the continuum, i.e. Gravity.

2.3.2 Reynolds Averaged Navier-Stokes (RANS)

Turbulent flow is a highly complex and chaotic phenomenon that is difficult to predict and analyse mathematically. Richard Feynman famously described turbulence as the most important unsolved problem in classical physics. [7]

When using the Navier-Stokes equations to describe fluid flow in computational fluid dynamics (CFD) simulations, there are challenges in accurately representing the intricate details of turbulent motion. Turbulent eddies can vary in size, from large to microscopically small, and to capture their effects fully, the computational grid needs to be extremely fine in specific areas. Direct numerical solution (DNS) of the Navier-Stokes equations without using any turbulence models requires approximately 10 to the power of 18 cells to capture and solve the smallest turbulent motions around the structure of a car [8], which is not considered practical due to the massive computing power required.

To address this computational challenge, turbulence models are often used to simplify the terms that describe turbulent motion in the Navier-Stokes equations. Two widely used methods of implementing turbulence models in CFD simulations are Reynolds-averaged Navier-Stokes (RANS) and Large Eddy Simulations (LES). Although both of these methods require less computational power than DNS, they cannot fully resolve the scales and interactions of turbulence, which makes it impossible to understand the physics of turbulence using these methods.

Despite its limitations, RANS provides a relatively computationally efficient way to model complex turbulent flows. However, the accuracy of RANS simulations depends heavily on the quality of the turbulence model used. Errors and uncertainties can be introduced due to the approximations made in the turbulence model, which can limit the predictive power of RANS simulations. Therefore, careful consideration and validation of the turbulence model are crucial to ensure accurate predictions of turbulent flow behaviour.

2.3.3 Turbulence Modelling

As mentioned in Reynolds Averaged Navier-Stokes (RANS), turbulence models are essential for accurate prediction of the turbulent flow behaviour. Some commonly used turbulence models are mentioned below.

2.3.3.1 K-Epsilon

K-Epsilon is a two-equation model that provides a computationally efficient way to account for the effects of turbulence in the Navier-Stokes equations. The K-Epsilon model is based on the concept of turbulent kinetic energy and its dissipation rate.

The model uses two transport equations, one for turbulent kinetic energy (K) and one for its dissipation rate (Epsilon). These equations are solved along with the continuity and momentum equations to determine the velocity and pressure fields in a turbulent flow.

The K-Epsilon model assumes that the turbulence scales are much smaller than the characteristic length scale of the flow and that the turbulence is isotropic. It also assumes that the turbulent viscosity is proportional to the turbulent kinetic energy and the dissipation

rate, and that the rate of dissipation of turbulent kinetic energy is proportional to the cube of the turbulent viscosity.

While the K-Epsilon model has its advantages in terms of computational efficiency, it also has its limitations. It is known to have difficulty accurately modeling certain types of flows, such as those with large-scale eddies or flow separation. [9]

2.3.3.2 K-Omega

In the K-omega model, like the K-epsilon model, two equations are solved simultaneously: the K equation and the omega equation. The K equation determines the turbulent kinetic energy, and the omega equation determines the specific dissipation rate of the turbulent kinetic energy.

One of the advantages of the K-omega model, unlike the K-epsilon model, is its ability to predict boundary layer behaviour accurately, especially near the walls. This is because the K-omega model includes a boundary layer correction term that helps to accurately predict the near-wall behaviour of turbulent flows.

However, like other turbulence models, the K-omega model is based on certain assumptions and simplifications, and there is always a trade-off between accuracy and computational efficiency. K-omega is known to produce unphysical results in certain flow situations, such as in highly separated flows or when there are strong streamline curvature effects and can have convergence issues. [9]

2.3.3.3 K-Omega SST (Shear Stress Transport)

K-omega SST is an extension of the K-omega model, which is designed to provide improved accuracy for wall-bounded turbulent flows, while maintaining computational efficiency.

The K-omega SST model is a hybrid model that combines the K-omega model and the K-epsilon model. It uses the K-omega model in the near-wall region, where viscous effects are dominant, and the K-epsilon model in the outer flow region, where the turbulent eddies are larger, and the effects of viscosity are less important.

The SST model is based on a transport equation for the turbulent kinetic energy (K) and the specific dissipation rate (omega). The K-omega SST model uses a blending function to smoothly transition between the K-omega and K-epsilon models, depending on the distance from the wall. This allows for accurate predictions of both the near-wall and far-field turbulence characteristics. K-omega SST is also considered to have less convergence issues than K-omega.

One of the main advantages of the K-omega SST model is that it is more accurate than the K-epsilon model for wall-bounded flows. However, like all turbulence models, the K-omega SST model has its limitations and can produce inaccurate results for certain flow situations and can still have convergence issues. [9]

2.4 Mesh

A mesh is a discretized representation of a computational domain that is used for numerical simulations. It is composed of a finite number of geometric elements which are combined to create a mesh that represents the domain.

The purpose of the mesh is to break down the governing equations, such as the Navier-Stokes equations, into small pieces so they can be solved more easily. The quality of the mesh can have a significant impact on the accuracy and efficiency of the numerical simulations.

One of the main factors that can have an impact on the result is its resolution or level of refinement. The mesh should be fine enough to capture the important flow features and variations in the solution, but not so fine that it becomes computationally infeasible to solve the equations.

Another important consideration is the type of mesh that is used. Structured meshes are regular and uniform, with a predictable and structured arrangement of elements. Unstructured meshes, on the other hand, can be more flexible and adaptable, allowing for more complex geometries and non-uniform element sizes. Some unstructured meshes are mentioned below.

2.4.1 Tetrahedral Mesh

Tetrahedral meshes are the simplest unstructured meshes and are widely used for geometries with relatively simple shapes. They are relatively easy to generate and offer good accuracy, but they can become computationally expensive for complex geometries. [10]

2.4.2 Polyhedral Mesh

Polyhedral meshes are more complex than tetrahedral meshes and are typically used for more complex geometries. They offer a good compromise between accuracy and computational efficiency, as they have fewer cells compared to tetrahedral meshes, while still capturing the geometric details of the simulation domain. However, they can be more challenging to generate and require more computational resources. [10]

2.4.3 Hexahedral Mesh

Hexahedral meshes or trimmed cells are used to model geometries with sharp edges or corners. They are essentially tetrahedral or polyhedral cells that have been trimmed to match the surface of the geometry. They offer good accuracy for sharp geometries but can require more computational resources than tetrahedral or polyhedral meshes. [10]

3 Design Concept

The Design concept is based on a multi-element wing with five individual elements. The central concept behind this design is to take a normal 3-element multi-element wing and split the main element into three smaller elements, thereby increasing the amount of air flowing underneath the wing and increasing the energy in the flow underneath the wing. This, if done correctly, can result in lower pressure under the wing, thereby increasing downforce and improving the vehicle's performance.

The five-element design of the wing allows for greater flexibility and control over the airflow around the wing. Each of the five elements can be adjusted independently to optimize the airflow. This greater control over the airflow allows for a higher potential for performance compared to a normal multi-element wing.

However, due to the increased complexity of the design, optimizing the performance of the concept will be more challenging. The design parameters for each of the five elements must be carefully considered and fine-tuned to achieve the desired performance gains.

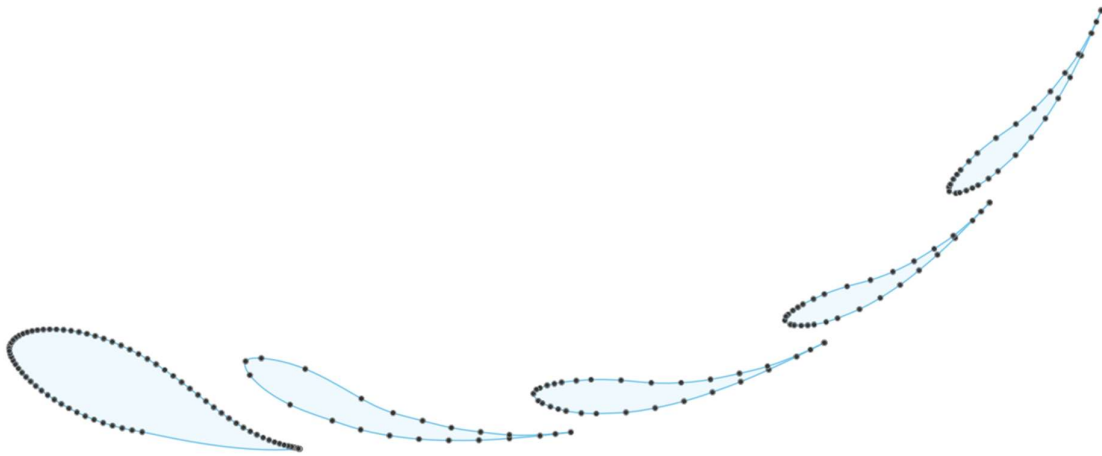


Figure 4. Picture visualizing text from Design Concept.

4 Methods

The design study of KTH Formula Student's vehicle, DeV18, utilises Siemens NX 2212 for 3D modelling and Simcenter Star-CCM+ 17.06.008-R8 for airflow simulations. A Design Manager Project inside Simcenter Star-CCM+ is used for optimization simulations. Design Manager Project allows for the user to specify a number of design parameters, such as angle of attack which the program then uses to run multiple simulations with varying design parameters to find the optimal design.

Smaller simulations are run on KTH Formula Student's own computers, while larger and more computationally heavy simulations are run on KTH's supercomputer, Dardel. This allows for faster processing times.

4.1.1 Multi-Element Wing Design

The design study for the multi-element wing is a process of iterative analysis with multiple simulation, to optimize its performance. The first step is to determine the most suitable airfoils and their positioning to each other through multiple Design Manager Projects using different airfoils and airfoil combinations in various positions relative to each other are used.

Due to limitations of 2D simulations, after the airfoils are selected and positioned relative to each other, simplified 3D simulations are run to estimate the optimal ground clearance and angle of attack for the whole wing. During the simplified 3D simulations, the airfoils are extruded to create a finite multi-element wing with flat endplates on the sides.

4.1.2 Side Structure Design 3D

After the Multi-Element Wing Design study, a well-performing multi-element wing for clean freestream airflow is apprehended. However, these conditions do not reflect the airflow the wing will be subject to once mounted on the car. Because of this, more complex 3D simulations are run, taking into account the rotating wheels and monocoque. The focus here is on cleaning up the airflow around the side structure to create conditions as similar as possible to the simplified 3D simulations used to optimize the multi-element wing. Key areas of focus are the tire squash and wake from the front wheels.

5 Simulation setup

5.1 Computational Domain and Boundary Conditions

The computational domain is the physical space that is divided into discrete cells or elements to perform calculations of the flow. The computational domain should be large enough to capture all the relevant features of the flow while also being small enough to avoid unnecessary computational expense. The domain is in this project defined by a council block as seen in Figure 5.

One important consideration when defining the computational domain is to ensure that the boundaries are far enough from the region of interest to minimize their influence on the flow. This is known as the boundary layer problem and can cause inaccuracies in the simulation if not properly addressed. In the automotive industry, a good size for the computational domain is between 5 and 20 car lengths downstream, behind the car and 2 to 3 car lengths in front and above the car. [11]

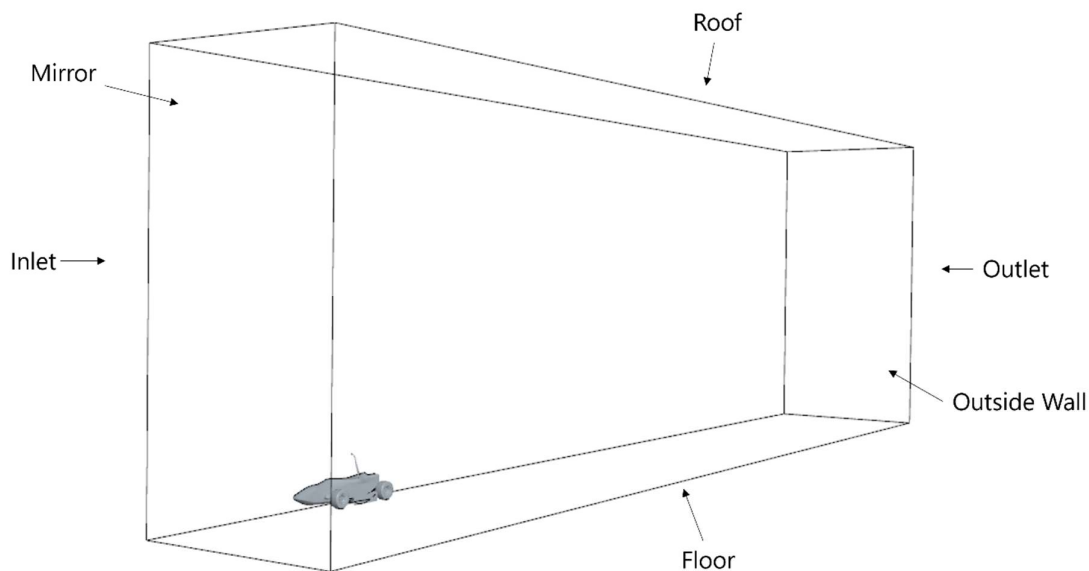


Figure 5. Picture showing the computational domain with named faces of the domain.

The size of the domain is selected to be 30 meters long, 15 meters high and 5 meters wide with 20 meters left of the domain behind the car.

To reduce computational time, a half-car simulation is set up for the symmetric car along its length axis. This is achieved by assigning a symmetry condition to the domain face that splits the car in half, as shown by the face "Mirror" in Figure 5. The face "Outside Wall" and "Roof" in Figure 5 are also assigned the boundary condition of symmetry as they are believed to have less impact on the results compared to a wall. The "Floor" face in Figure 5 is assigned a boundary condition of no-slip wall with a tangential velocity downstream of 40 kph to simulate the ground the car is travelling on. The "Inlet" and "Outlet" faces in Figure 5 are assigned the boundary conditions of velocity inlet with a value of 40 kph and pressure outlet with a value of 0 Pa, respectively. All boundaries on the car are assigned the boundary condition no-slip wall with the wheels also being assigned a rotational velocity condition with a value of 480.1046... rpm which would replicate DeV18 wheels rotating when the car is travelling at a velocity of 40 kph.

5.2 Mesh Setup

The mesh used for all simulations can be found in Table 1 and Table 2.

Table 1. Mesh setup.

Mesh Properties	Selected Option
Mesher	Polyhedral Mesher
Prism Layer Mesher	Yes
Base Size	0.1 meters
Target Surface Size	100% of Base
Minimum Surface Size	10% of Base
Surface Curvature	150.0 pts/circle
Number of Prism Layers	10
Prism Layer Stretching	1.4
Prism Layer Total Thickness	10% of Base
Volume Growth Rate	1.2
Surface Growth Rate	1.3

Volumetric controls and surface controls are used to further refine the mesh around areas of interest.

Table 2. Custom controls.

Volumetric controls	Controls	Values
Leading and Trailing Edge	Custom Size	1% of Base
Wake	Custom Size	5% of Base
Surface Controls	Controls	Values
Wing Element	Target Surface Size	2.5% of Base
	Surface Curvature	540 pts/circle

5.3 Physics Setup

The physics set up used for all simulations can be found in Table 3.

Table 3. Physics setup.

Physical Properties	Selected Option
---------------------	-----------------

Space	Three Dimensional
Material	Gas
Flow	Segregated Flow
Equation of State	Constant Density
Time	Steady
Viscous Regime	Turbulent
Reynolds-Averaged Turbulence	K-omega Turbulence
K-omega Turbulence Model	K-omega SST
Wall	All y^+ Wall Treatment

6 Results

6.1 Multi-Element Wing Design

See Figure 6 for results from 2D-simulations where the goal was to decide on airfoils as well as their positioning in respects to each other,

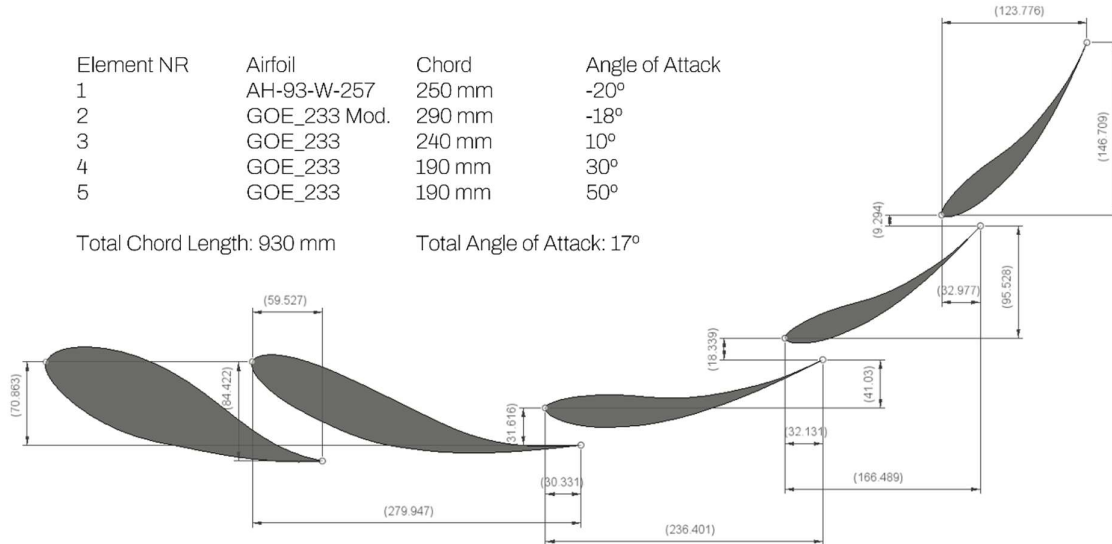


Figure 6. Results from 2D simulations of Multi-Element Wing Design.

Figure 7 shows results from design manager project for simplified 3D-simulations where design parameters total ground clearance and total angle of attack was used.

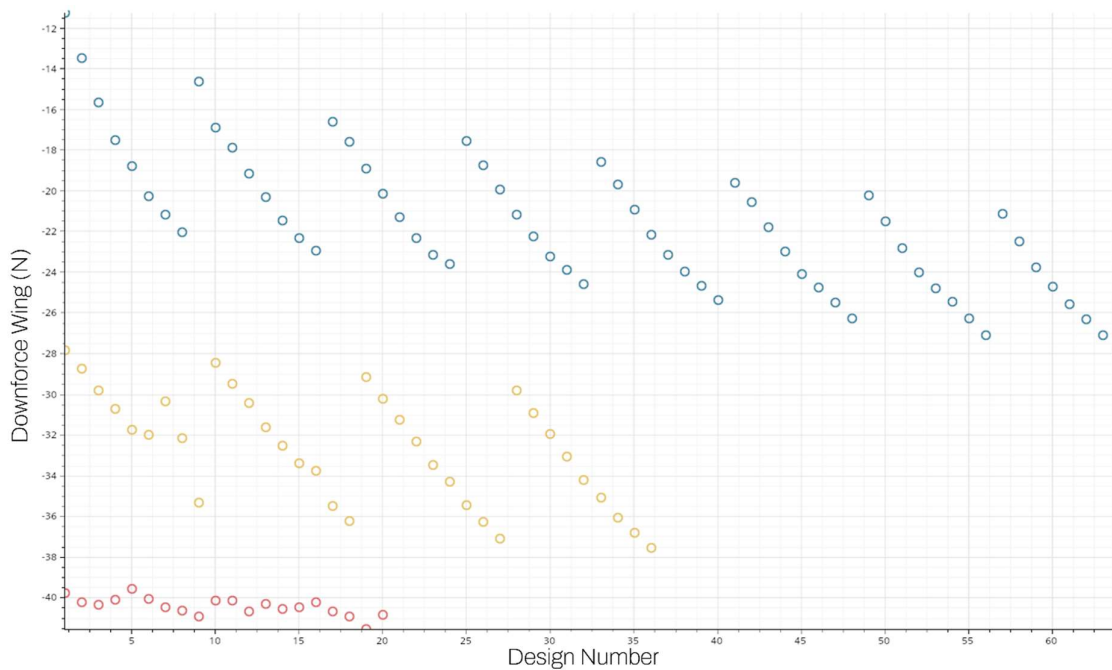


Figure 7. Result from ground clearance and angle of attack study of Multi-Element Wing Design.

A total of 128 simulation were run, where total ground clearance could vary from 50 mm to 300 mm and total angle of attack from 12° to 25°. Overall, best design had a ground clearance of 270 mm and angle of attack of 25°.

6.2 Side Structure Design 3D

Multiple designs were simulated to try and control the wake and tire squash from the front wheels with the most effective design shown in Figure 8 and Figure 9. See also Appendix 1 for more pictures.



Figure 8. Close look on results from Side Structure Design 3D.

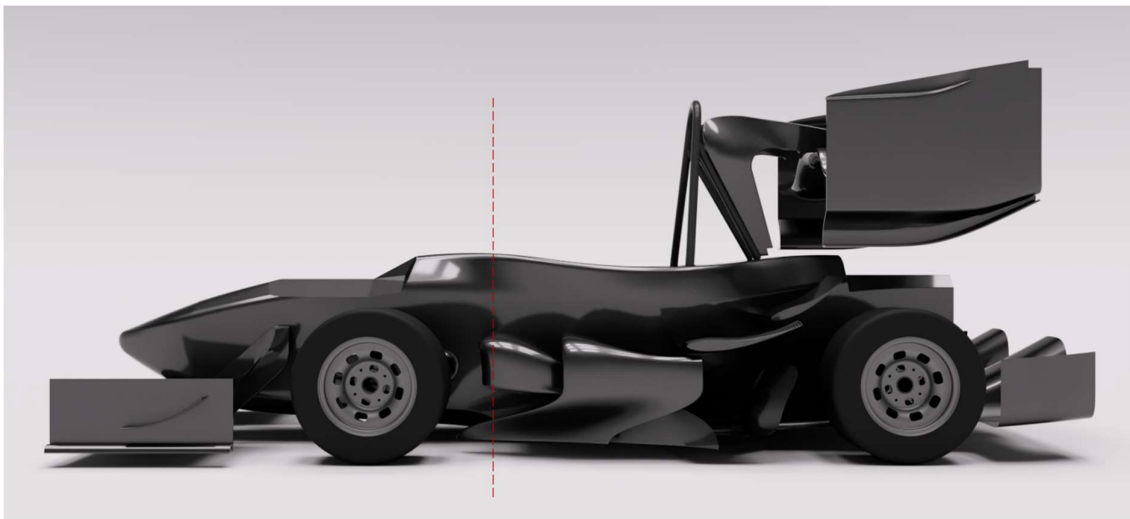


Figure 9. Side View of DeV18 with results from Side Structure Design 3D.

The idea behind this concept it to let the first two element curve down to try and keep the tire wake and squash away from underneath the wing. The second element then also curves in front of the first element to act as the main barge board with the first element acting behind to outwash any potential spill under. As seen on the left side in Figure 10, this concept does a good job at keeping low energy air (blue colour) away from the wing.

The resulting side structure produces a total of 26 N of downforce and 6N of drag at 40 kph.

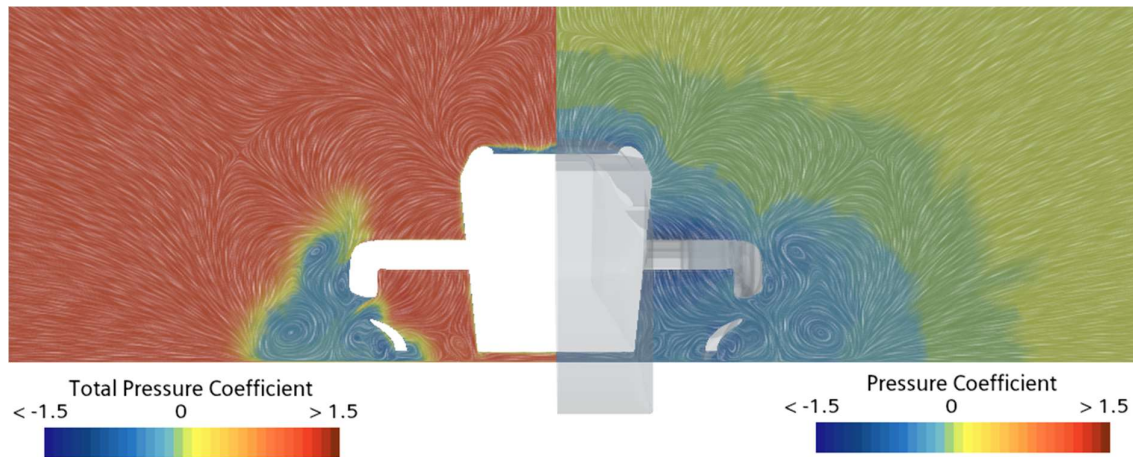


Figure 10. Cut plot of Total Pressure Coefficient and Pressure Coefficient at red dotted line in Figure 9.

7 Discussion

7.1 Concept Discussion

The design apprehended outperforms the side structure of the DeV17 by quite a lot in pure downforce figures. The multi-element wing design produces more than two times the downforce when the car is travelling at 40 kph while still producing less drag.

Although the design outperformance it is believed that, in order to compete with the top teams, further increase in performance is required. To achieve this, a more complex optimization method of the multi-element wing could be used. Such as an adjoint optimization method.

The design apprehended outperforms the side structure of the DeV17, producing more than two times the downforce at 40 kph while still generating less drag. However, in order to compete with top teams, it is believed further improvements in the side structure are necessary.

One potential method for achieving these improvements is through the use of an adjoint optimization technique. This approach involves the calculation of gradients of the objective function with respect to the design variables using the adjoint method, which provides a more computationally efficient way to obtain optimal designs.

The use of this method could undoubtedly improve the pressure distribution under the wing further increasing performance but there are still question marks on whether that would be enough to reach the desired performance goals. It is believed that other design concepts such as a venturi-based floor could reach higher, or at least the same, amounts of downforce whilst being more drag efficient and more compatible with the rear wing. See Appendix 2.

7.2 Accuracy

A crucial aspect of any CFD study is understanding the weaknesses in the model and how they affect the results. One key area that can have a big impact on the results is the mesh. To ensure good enough mesh quality, one can perform a mesh convergence study, also known

as a mesh independence study. The objective of this study is to refine the mesh to make sure that the results are not dependent on the grid size.

The convergence study is conducted by first selecting an initial mesh that is coarse enough to run the simulation quickly yet detailed enough to capture essential features of the geometry and flow field. The mesh is set up with a changeable global parameter that increases the mesh density in the near field (flow field near the car) and wake with an increased value. Multiple iterations of the same simulation are then conducted with progressively finer mesh until the results no longer significantly change with respect to the mesh refinement.

The mesh used throughout this project is based on the DeV17 full car simulation model, for which a mesh convergence analysis was conducted. This should mean good mesh quality for the results from this study, although a new mesh convergence study would be beneficial to ensure minimum mesh influence.

Another area that can have a drastic impact on the results is the choice of turbulence model, initial conditions, and under-relaxation factors. Under-relaxation factors are used to control the rate at which the solution variables are updated during each iteration of the solution process. Setting the under-relaxation factors too high can lead to instability and slow convergence, while setting them too low can lead to slow convergence and excessive computational cost. Compared to the mesh, the choice of these is more difficult to make since they require comparison to the real world to validate. Because of this, this project has heavily depended on literature.

7.3 Future Work

Although further improvements of the concept are possible, this concept will no longer be investigated since it is believed the overall potential of the concept is too small. Instead, another concept will be investigated, such as the one seen in Appendix 2.

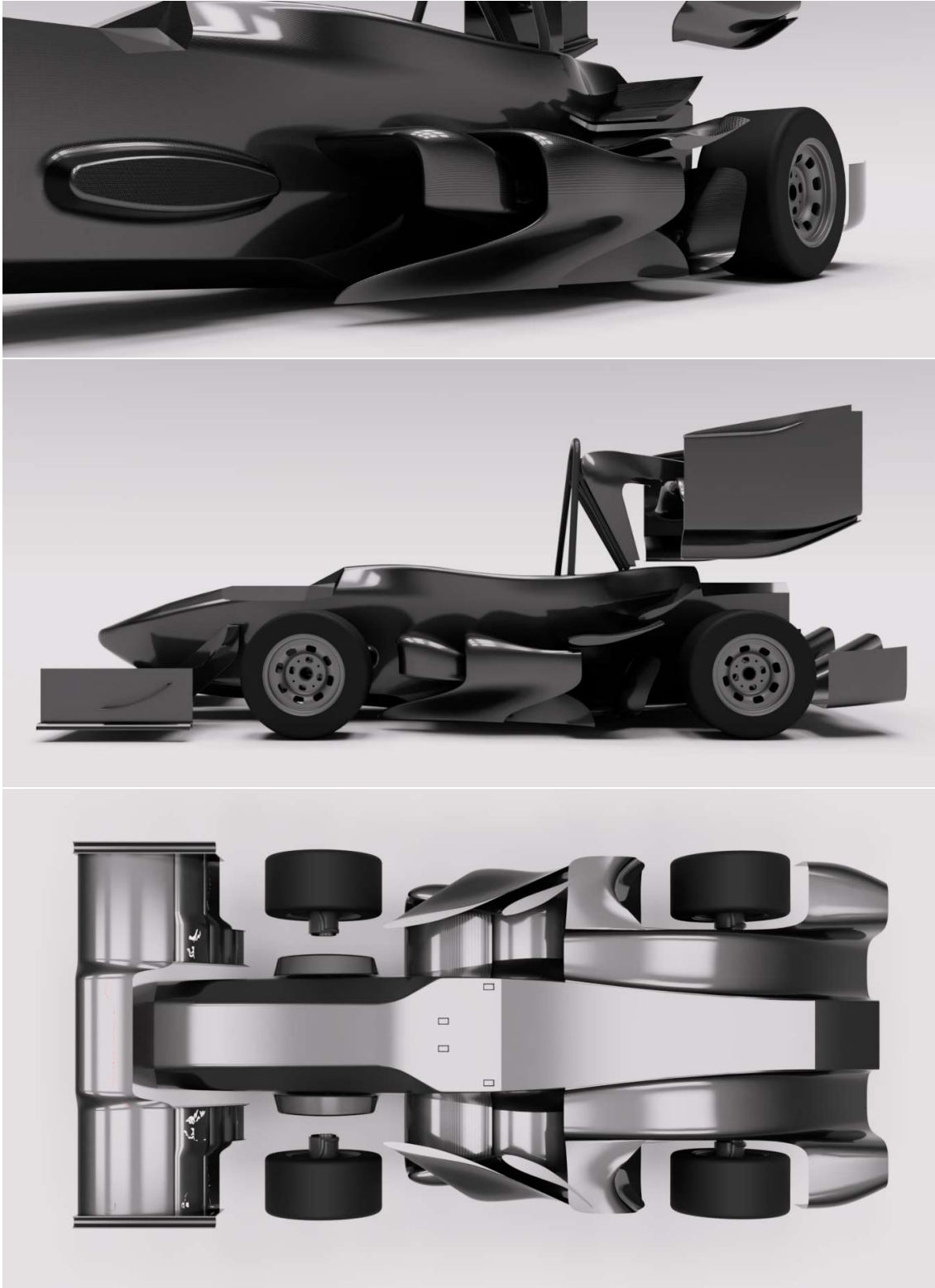
Additionally, a new mesh convergence study will be conducted as well as visiting a wind tunnel with the DeV17 to validate the choice of turbulence model, initial conditions, and under-relaxation factors. By doing this, future simulations will be more accurate which, in the long term, will allow for better results at competition.

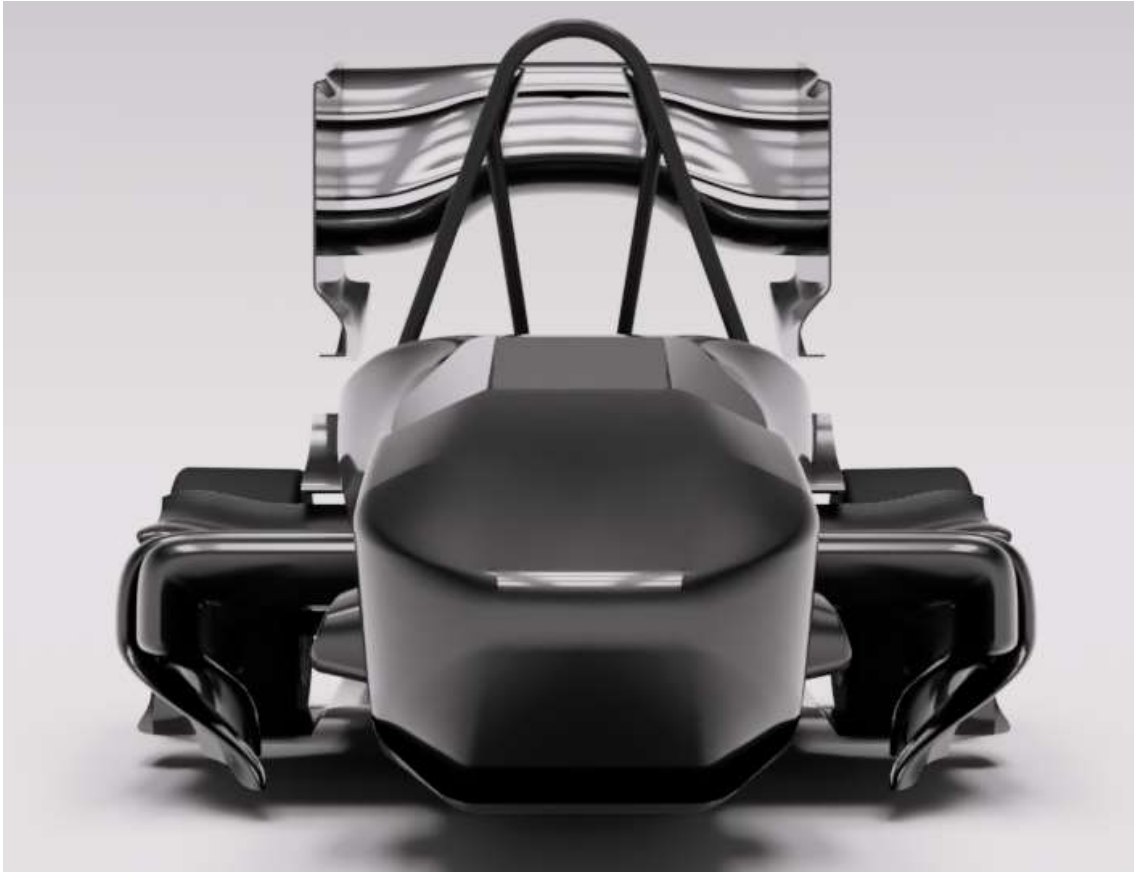
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Appendix 1

Pictures of results from Side Structure Design 3D





Appendix 2

